

1. Project Overview

Project Title: Vendor Performance Analysis

Industry Context: Retail & Wholesale Distribution

Project Goal: To analyze vendor performance across sales, profitability, inventory efficiency, and pricing strategy, and provide actionable recommendations for optimizing business outcomes.

Key Focus Areas:

- Vendor profitability and sales contribution
- Inventory turnover and holding costs
- Pricing efficiency (bulk purchasing impact)
- Brand-level underperformance identification
- Performance gap analysis between high and low-performing vendors

2. Business Problem Definition

Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. Companies need to ensure that they are not incurring losses due to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to:

- Identify underperforming brands that require promotional or pricing adjustments.
- Determine top vendors contributing to sales and gross profit.
- Analyze the impact of bulk purchasing on unit costs.
- Assess inventory turnover to reduce holding costs and improve efficiency.
- Investigate the profitability variance between high-performing and low-performing vendors.

3. Dataset Description

The analysis uses 6 CSV files containing transactional and reference data for vendor performance analysis. Total data volume exceeds 15 million rows.

#	Table Name	Row Count	Description
1	begin_inventory	206,529	Opening inventory snapshot
2	sales	12,825,363	Sales transactions by vendor and brand
3	purchase_prices	12,261	Product pricing reference data

4	purchases	2,372,474	Purchase orders from vendors
5	vendor_invoice	5,543	Freight and invoice data
6	end_inventory	224,489	Closing inventory snapshot

Total Records: ~15.6 million rows

Primary Join Keys:

- VendorNumber / VendorNo – Links vendor-related tables
- Brand – Links product pricing and transaction tables

4. Tools & Workflow

Phase	Key Activities	Tools Used
1	Business understanding, table exploration	SSMS, Excel
2	Load CSV files into the database	Python, SQLAlchemy
3	Clean data, handle NULLs, remove duplicates	SQL (SSMS)
4	Merge tables, create an aggregated table, and calculate metrics	SQL (CTEs, Joins)
5	Summary stats, distributions, correlations	Python (Pandas, Matplotlib, Seaborn)
6	T-tests, confidence intervals, and correlation	Python (SciPy, NumPy)
7	Answer 5 core business questions	SQL + Python
8	Interactive dashboard with slicers	Power BI
9	Synthesize findings, prioritize actions	Report
10	Documentation, GitHub upload	Markdown, Git

5. EDA (Python) and SQL Analysis

In this phase, I examined the various tables in the database to identify key variables, understand their relationships, and determine which ones should be included in the final analysis

1. Created a database and imported the CSVs into the database in SQL Server
2. Explored the database
 - Listed all tables and checked row counts
 - Previewed data to understand columns and relationships
3. Identified how tables connect
 - VendorNumber and Brand were the key links
 - Purchases, sales, freight, and pricing data were spread across multiple tables
4. Created a master table using SQL CTEs
 - Combined purchases, sales, freight, and purchase prices
 - Grouped by vendor and brand
 - Result: One table called vendor_sales_summary
5. Cleaned the data in Python
 - Fixed data types, filled missing values, removed whitespace
 - Replaced infinite values with 0
6. Added calculated metrics
 - Gross Profit, Profit Margin, Stock Turnover, Sales-to-Purchase Ratio
7. Saved back to SQL Server
 - Created a new table for faster dashboarding and reporting and saved back in SQL server

6. Business Analysis & Statistical Findings

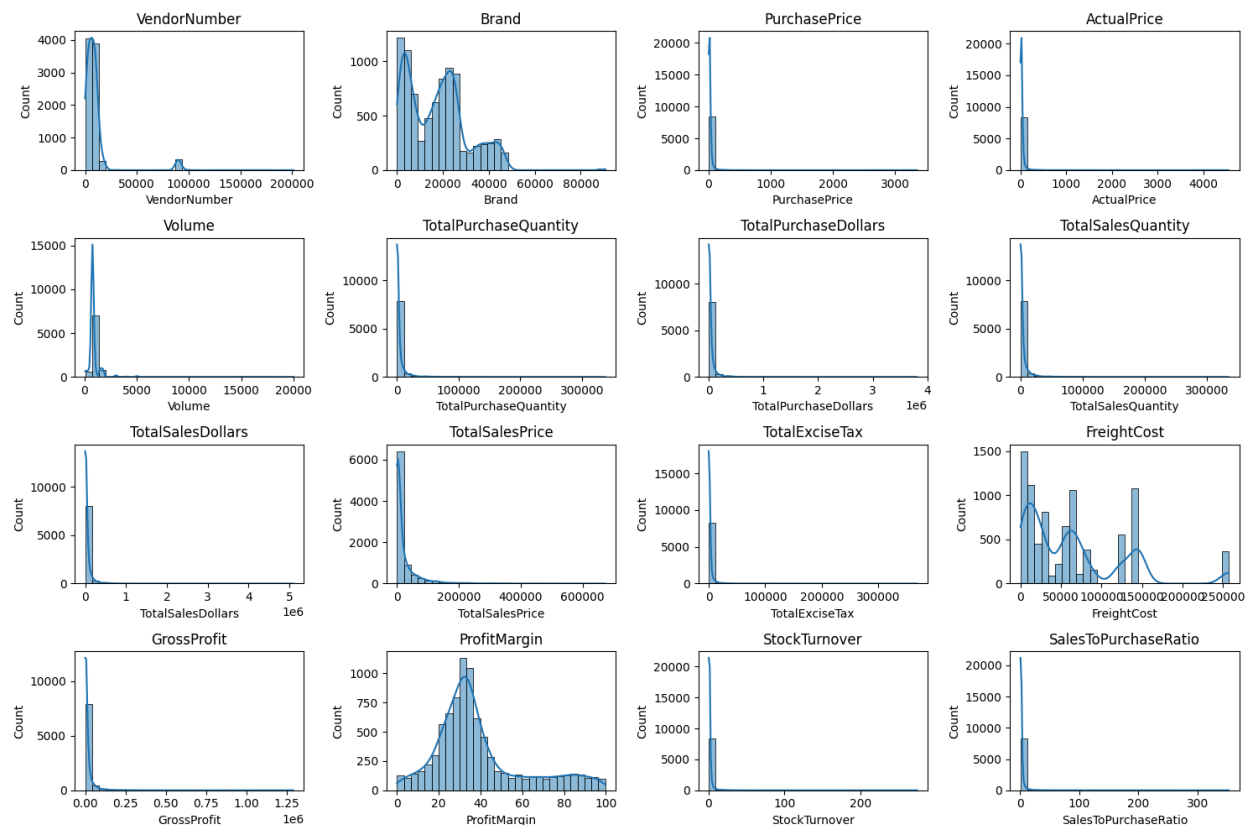
6.1 Summary Statistics

I generated summary statistics for all numerical columns to understand data distribution, central tendencies, and potential issues.

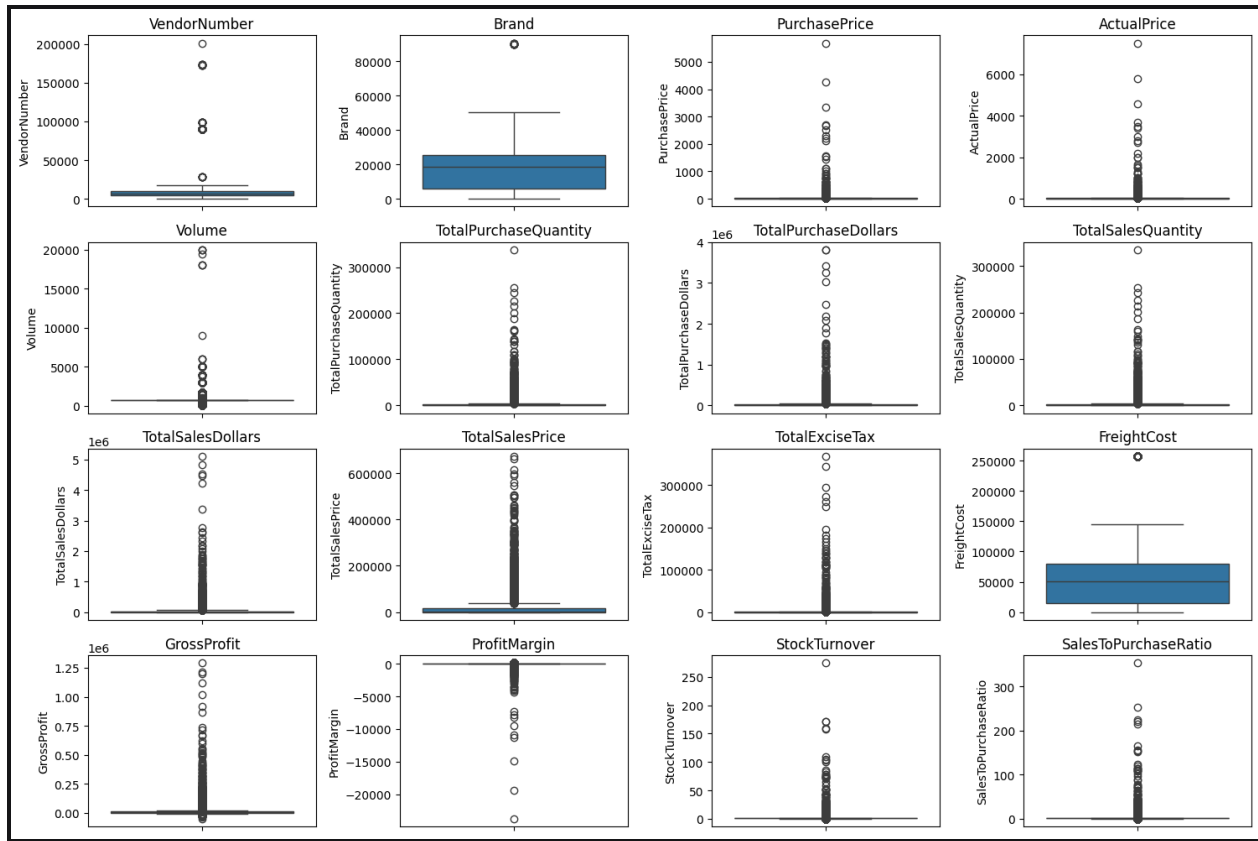
	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	10650.649458	18753.519148	2.000000	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	18039.228769	12662.187074	58.000000	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	24.385303	109.269375	0.360000	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	35.643671	148.246016	0.490000	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	847.360550	664.309212	50.000000	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3140.886831	11095.086769	1.000000	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	30106.693372	123067.799627	0.710000	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3077.482136	10952.851391	0.000000	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	42239.074419	167655.265984	0.000000	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	18793.783627	44952.773386	0.000000	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1774.226259	10975.582240	0.000000	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	61433.763214	60938.458032	0.090000	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	12132.381048	46224.337964	-52002.780000	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-15.620770	443.555329	-23730.638953	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793	6.020460	0.000000	0.807229	0.981529	1.039342	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504390	8.459067	0.000000	1.153729	1.436894	1.665449	3.529286e+02

6.2 Distribution & Outlier Analysis

Distribution Plots: Created histograms with KDE curves for all numerical columns to visualize data spread.



Boxplots: Used boxplots to detect outliers in each numerical column.

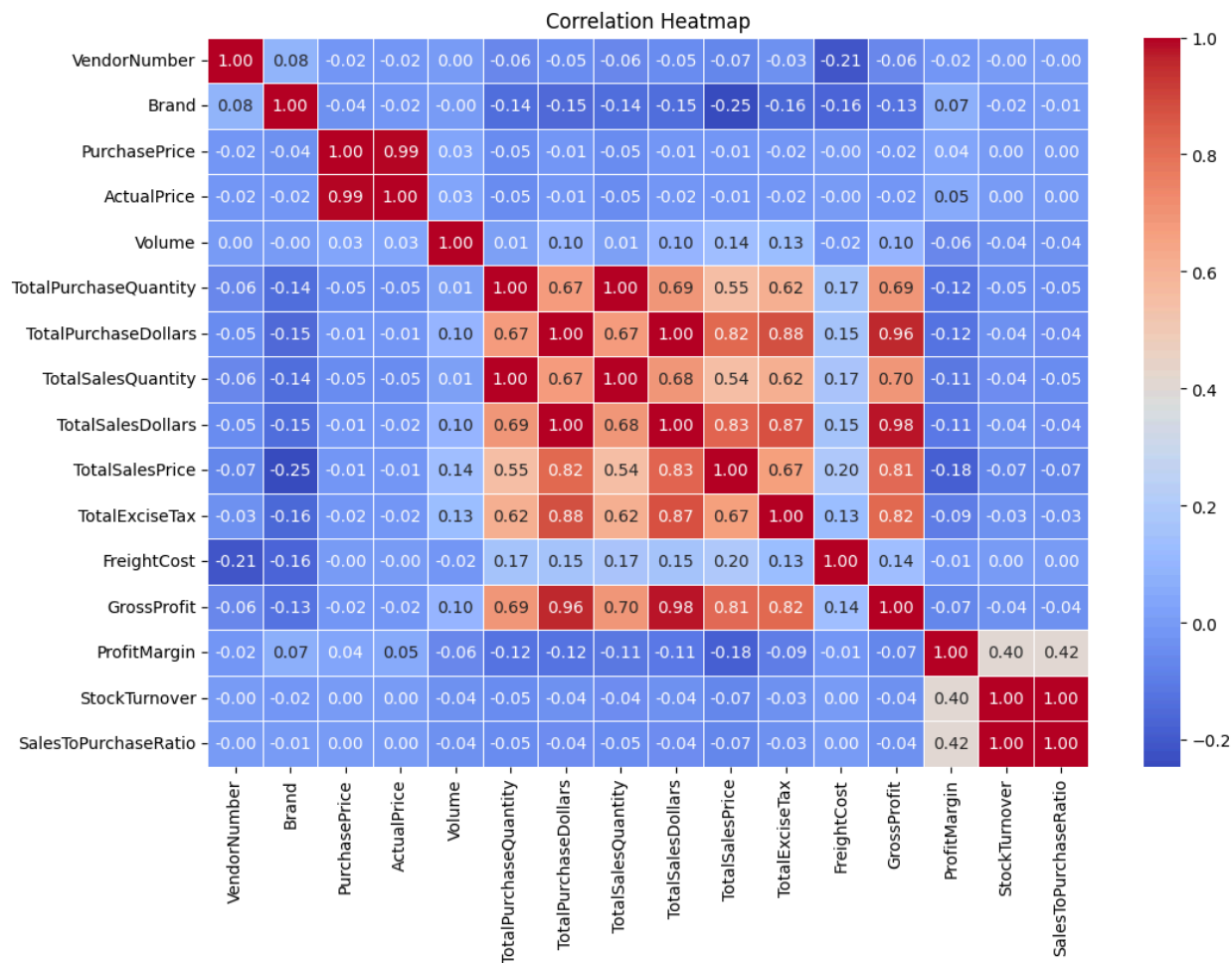


6.3 Key Findings from Initial EDA

Issue	Finding
Negative Gross Profit	Minimum value -\$52K → some products are selling at a loss
Zero Sales	Products purchased but never sold → slow-moving or obsolete stock
High Price Variation	Purchase price max 5,682 vs mean 24 → premium products exist
Freight Cost Variation	Ranges from 0.09 to 257K → logistics inefficiencies
Stock Turnover Range	0 to 274.5 → some products sell extremely fast, others sit idle

6.5 Correlation Analysis

Created a correlation heatmap to understand relationships between key metrics.

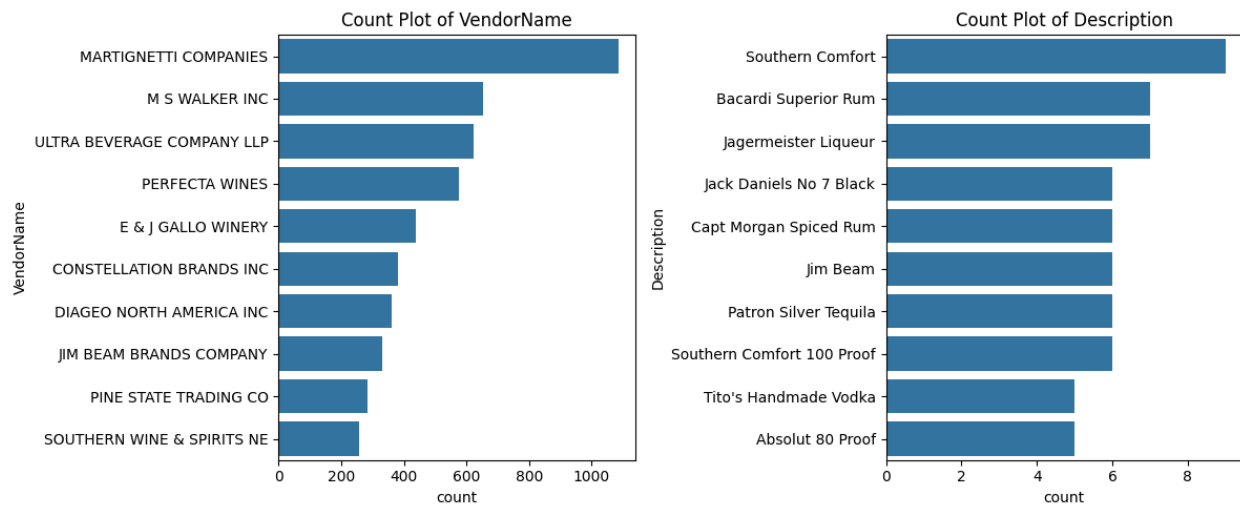


Key Correlations Found:

Relationship	Correlation	Insight
Purchase Quantity vs Sales Quantity	1.00	Efficient inventory turnover
Stock Turnover vs Profit Margin	0.40	Faster-selling inventory = higher margins
Stock Turnover vs Gross Profit	-0.04	Turnover doesn't predict overall profit
Profit Margin vs Sales Price	-0.18	Higher prices may reduce margins due to competition

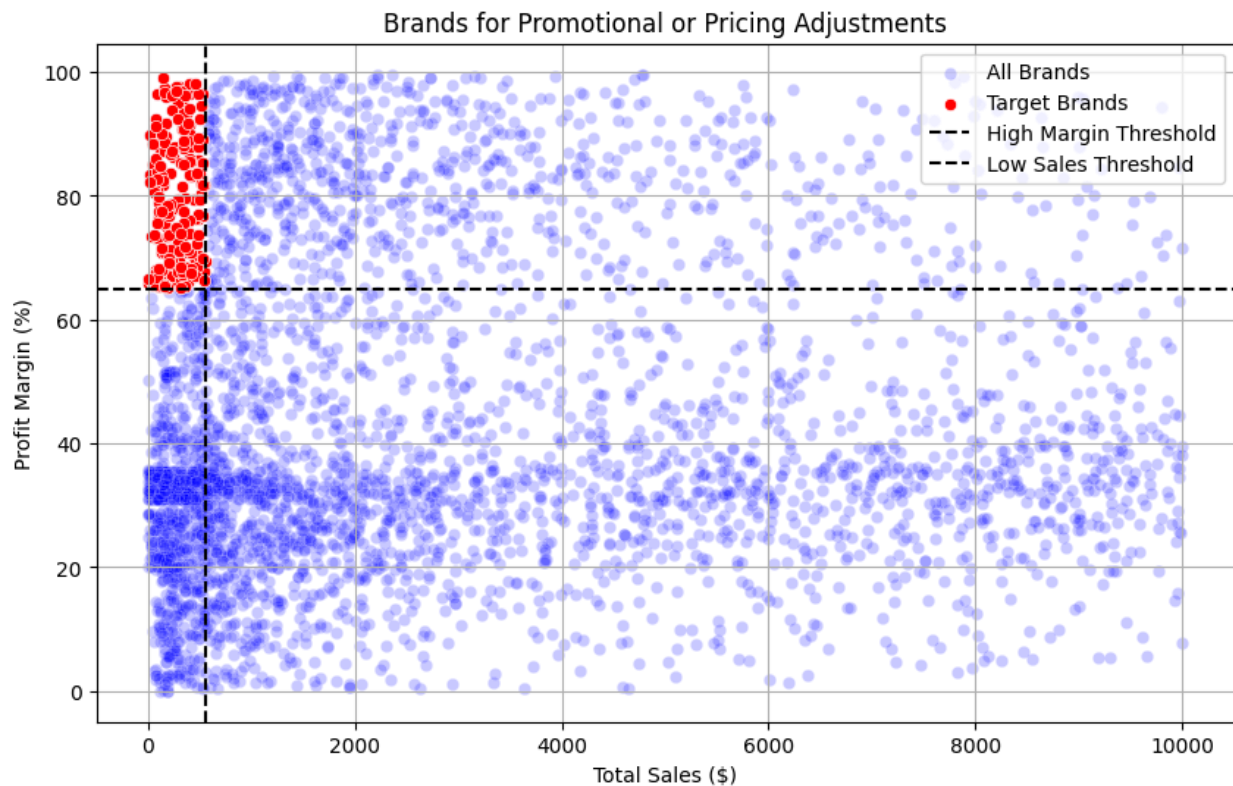
6.6 Count Plots for Categorical Columns

Visualized the top 10 vendors and products by frequency.



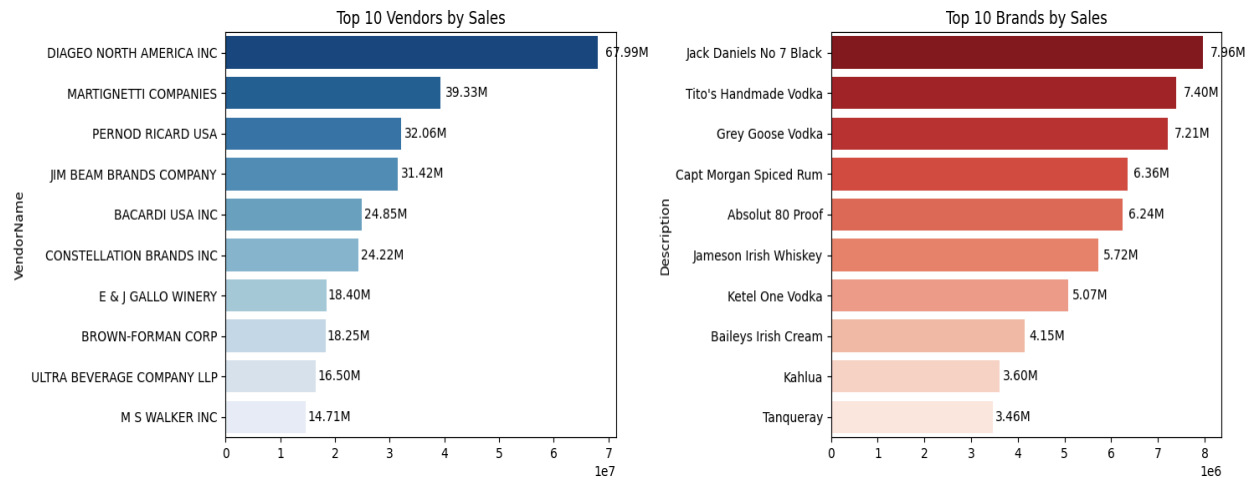
6.7 Brands Needing Promotional or Pricing Adjustments

Approach: Identified brands with low sales (bottom 15%) but high profit margins (top 85%).



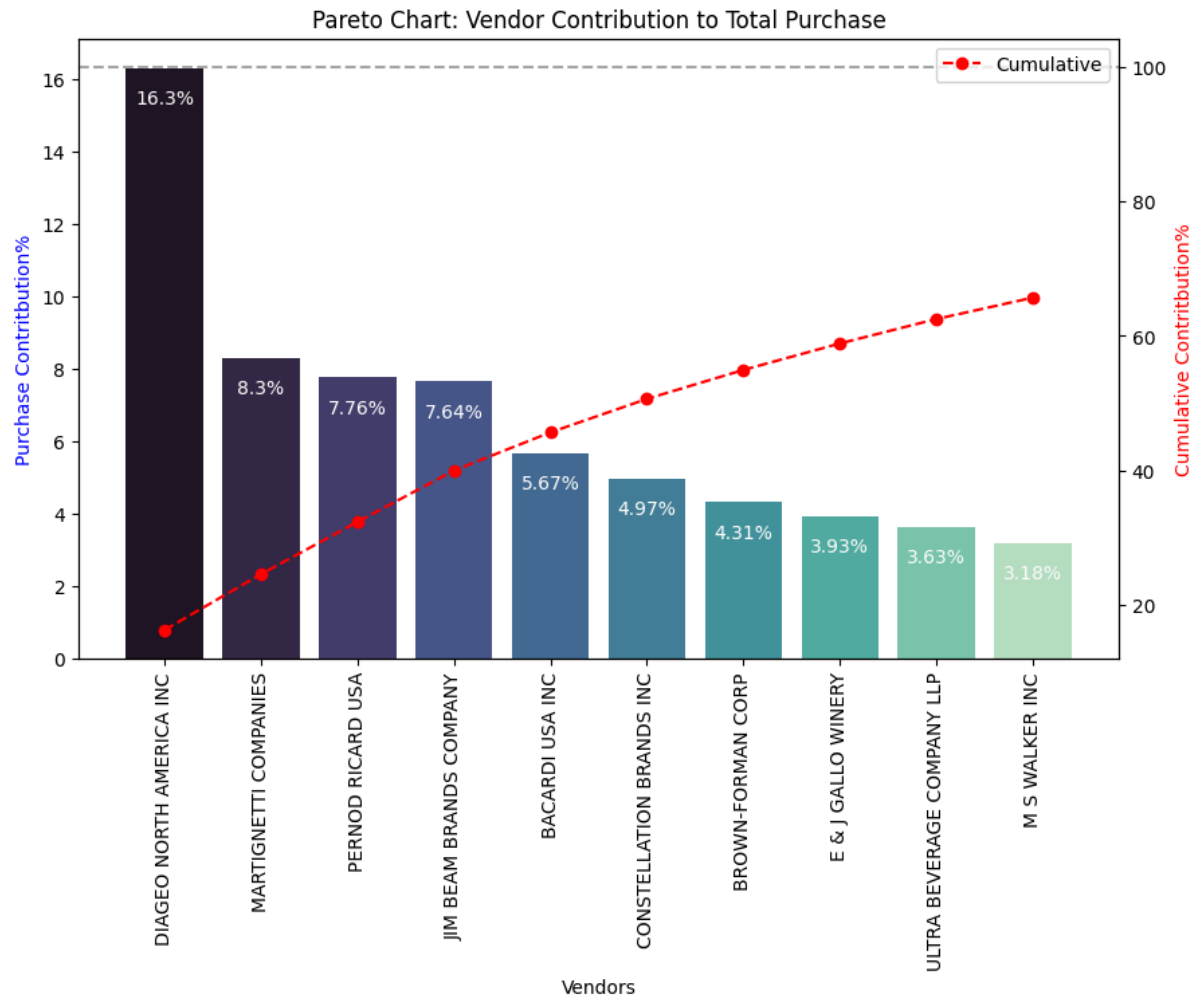
Result: Brands with low sales volume but high margins → prime candidates for promotion to drive volume without sacrificing profitability.

6.8 Top Performers: Vendors & Brands by Sales



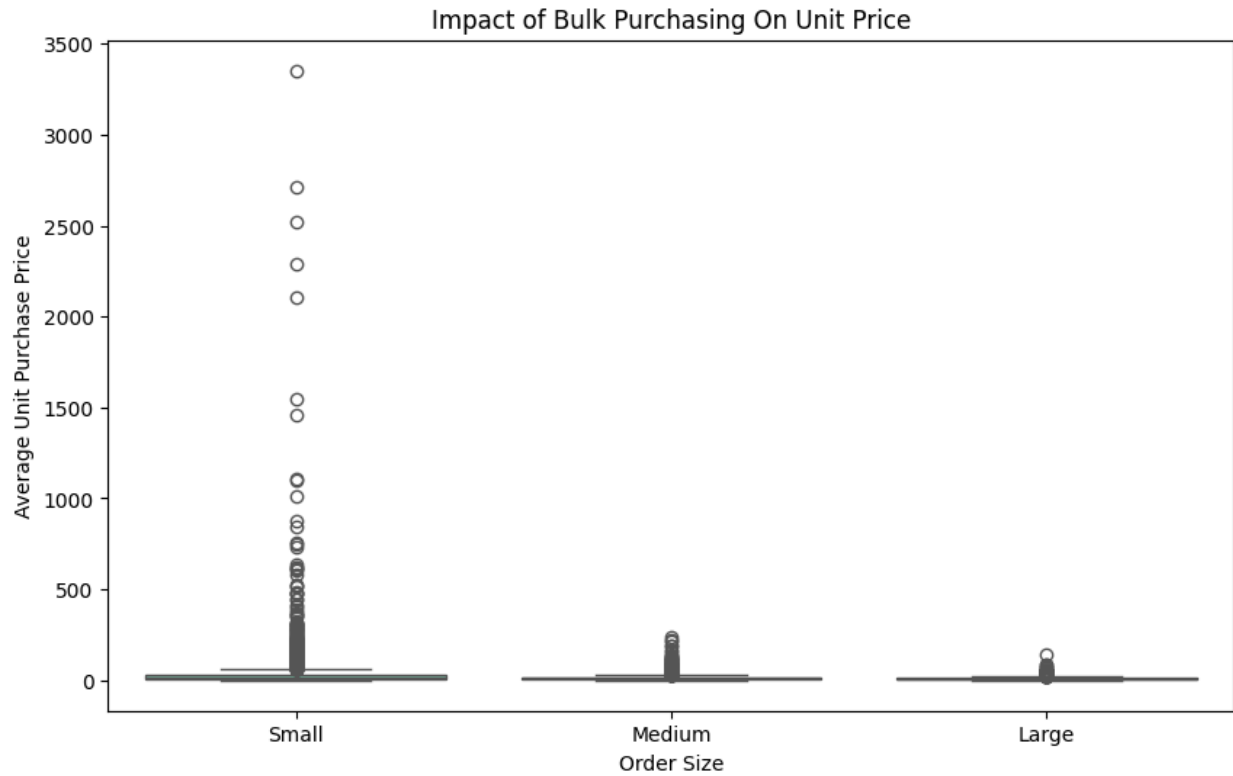
6.9 Purchase Contribution & Pareto Analysis

Findings: Top 10 vendors contribute 65.69% of total purchase dollars.



6.10 Bulk Purchasing Impact on Unit Price

Analysis: Segmented orders into Small, Medium, Large based on purchase quantity.



Order Size	Average Unit Price
Small	Higher
Medium	Medium
Large	\$10.78 (lowest)
Insight: Large orders achieve ~72% cost reduction compared to small orders.	

6.11 Inventory Efficiency

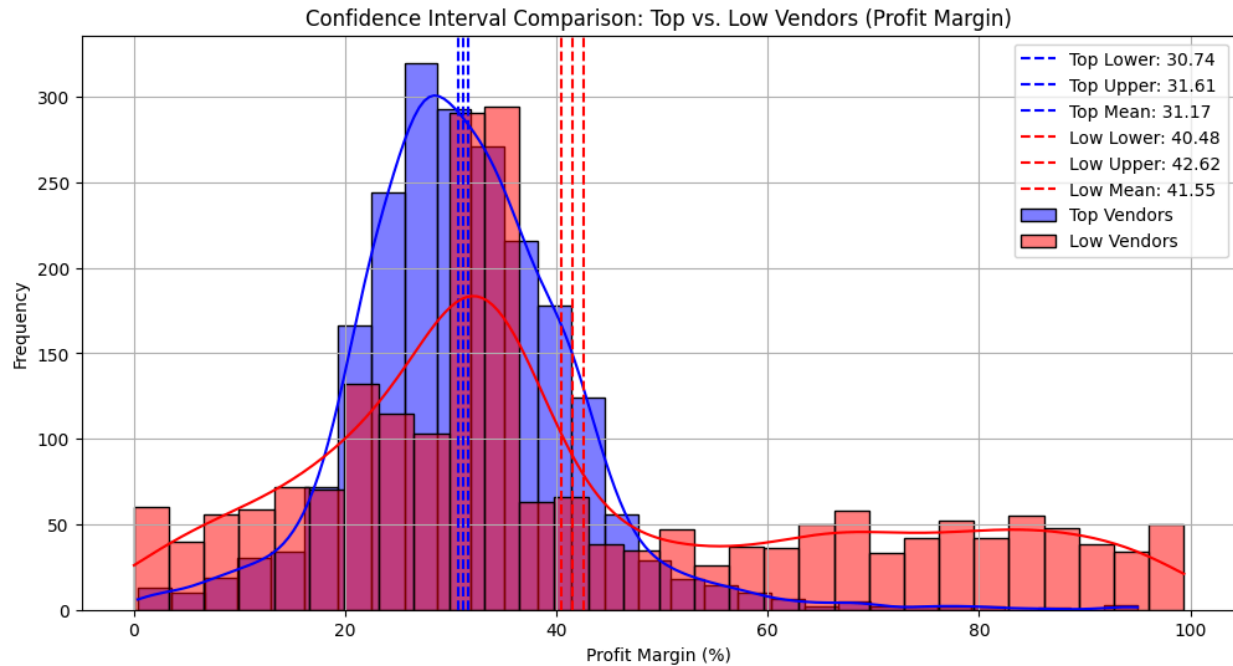
Low Stock Turnover Vendors: Identified vendors with turnover < 1 (excess stock).

	VendorName	UnsoldInventoryValue	UnsoldinventoryValue
25	DIAGEO NORTH AMERICA INC	722209.05	722.20905K
46	JIM BEAM BRANDS COMPANY	554665.63	554.66563K
68	PERNOD RICARD USA	470625.61	470.62561K
116	WILLIAM GRANT & SONS INC	401960.83	401.960830000000004K
30	E & J GALLO WINERY	228282.61	228.28261K
79	SAZERAC CO INC	198436.41	198.43641K
11	BROWN-FORMAN CORP	177733.74	177.733739999999998K
20	CONSTELLATION BRANDS INC	133617.62	133.61762K
61	MOET HENNESSY USA INC	126477.70	126.477699999999998K
77	REMY COINTREAU USA INC	118598.15	118.598149999999999K

6.12 Statistical Analysis: Confidence Intervals & T-Test

95% Confidence Intervals for Profit Margin:

Vendor Type	Confidence Interval	Mean
Low-Performing	40.48% – 42.62%	~41.5%
Top-Performing	30.74% – 31.61%	~31.2%



T-Test Results:

- T-Statistic: -17.64
- P-Value: 0.0000
- Conclusion: Reject Null Hypothesis → Significant difference in profit margins exists between top and low-performing vendors.

Key Insight: Low-sales vendors maintain higher profit margins (premium pricing), while high-sales vendors operate on thinner margins (volume strategy).

7. Power BI Dashboard

The Power BI dashboard provides an interactive overview of vendor performance, profitability, and inventory efficiency. Key metrics at the top show total sales of 441.41M, total purchases of 307.34M, gross profit of 134.07M, Profit Margin of 38.72, and unsold capital worth of 2.71M.

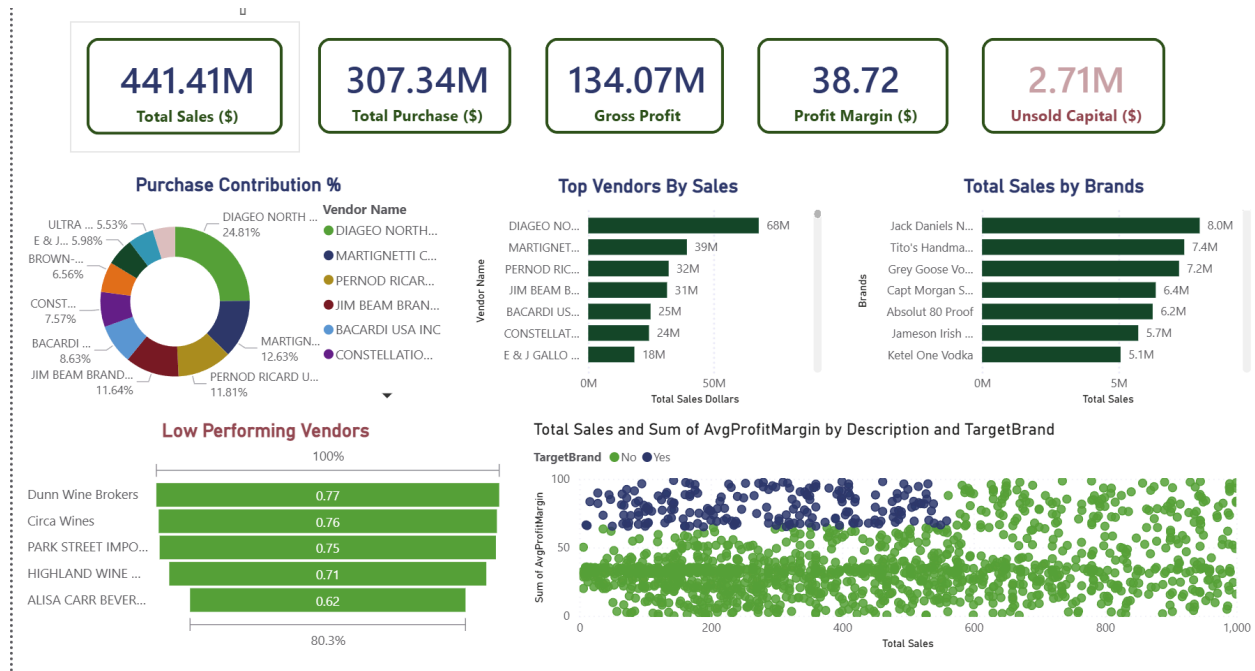
These high-level KPIs give stakeholders an immediate snapshot of business health.

The dashboard includes several visualizations to support decision-making:

- A Pareto chart showing vendor purchase contribution with cumulative line (top 10 vendors account for ~65.69% of total purchases)
- Bar charts for the top 10 vendors and the top 10 brands by sales
- A scatter plot illustrating the 0.40 correlation between stock turnover and profit margin
- A confidence interval comparison chart showing low-performing vendors have higher profit margins (40.48%–42.62%) than top performers (30.74%–31.61%)

- Slicers for filtering by vendor name, brand, and product category

Users can click on any visual element to cross-filter the entire dashboard, enabling dynamic exploration of vendor and brand performance.



8. Recommendations

Based on the analysis of vendor performance, profitability, inventory turnover, and statistical testing, the following recommendations are proposed:

- For High-Performing Vendors (High Sales, Lower Margins)
 - Optimize pricing strategy – Test small price increases (2-5%) on top-selling products to improve margins without significant volume loss
 - Reduce operational costs – Negotiate better freight rates or consolidate shipments to lower cost of goods sold
 - Bundle products – Encourage higher average order value through product bundling to increase gross profit per transaction
- For Low-Performing Vendors (Low Sales, Higher Margins)
 - Increase marketing support – Promote high-margin products through targeted campaigns to drive sales volume
 - Expand distribution channels – Make these products available across more stores or platforms
 - Offer volume discounts – Temporarily reduce prices to attract customers while still maintaining healthy margins

3. Inventory Management

- Address slow-moving stock – Review products with stock turnover below 1. Consider markdowns, promotions, or discontinuation
- Reduce unsold capital – \$2.71M is tied up in unsold inventory. Implement better demand forecasting to align purchases with sales
- Focus on bulk purchasing – Large orders achieve ~72% lower unit cost. Incentivize vendors to buy in bulk where demand is predictable

4. Vendor Strategy

- Strengthen relationships with top vendors – Top 10 vendors contribute 65.69% of purchases. Prioritize relationship management and negotiate better terms
- Help low-sales, high-margin vendors grow – Collaborate with these vendors to increase their sales volume while protecting their margins

5. Promotional Strategy

- Target brands with low sales but high margins – These are ideal candidates for promotions to drive volume without eroding profitability
- Monitor discount dependency – Ensure discounts are driving incremental sales, not just shifting demand